



Smart Electronics, Smarter Factories

- Electronics fuel smart factories; smart manufacturing, and, in turn, drive electronics innovation
- Industry growth: 10% globally, even faster in India
- Roadblocks: Shorter product cycles, price wars, old machines, scrap losses, skill gaps, and sustainability demands
- The shift: Connect→Integrate→AI-power
- True success=process+technology+people working in sync

ERP, and MES, and their integration, must be deployed horizontally across the factory, including different departments and functions. While some technologies can be planned on a point-to-point basis, for instance, automating manual processes, material movement, and quality inspection using robots, cobots, AMRs (autonomous mobile robots), and AI-computer vision, it is vital to assess which processes can be digitised, focusing on whether they add value in OpEx reduction and improve quality and safety.

Factory health and performance

To maintain factory performance, industries must track the performance of each machine and piece of equipment, particularly those located at critical stations. Checking key metrics such as availability, performance, and quality (APQ) is vital for achieving overall equipment effectiveness.

Moving towards smart manufacturing environments, this is especially relevant for the electronics sector, which plays a dual role: it transforms internally and enables transformation in other industries, such as automotive, aerospace, and industrial manufacturing. This dual role makes it essential to evaluate the electronics industry from both perspectives.

Challenges of the electronic manufacturing industry

• **Product life cycles.** These have become significantly shorter. Automotive programs that once lasted four to five years are now replaced in one to two years. In the consumer electronics industry, constant demand for new features pressures manufacturers

to innovate rapidly.

• **Pricing pressures.** Competition is intense. Managing costs—not just the final product price but also CapEx and OpEx—is crucial for sustaining profitability.

• **Brownfield machines.** Factories often have many existing machines that are non-communicating with each other. Replacing old machines is neither feasible nor cost-effective, so retrofitting with IoT devices, sensors, or similar upgrades is necessary to enable meaningful analytics.

• **Scrap management.** Due to critical operations and precision manufacturing requirements, quality rejections are very high, leading to heavy process scrap.

• **Skilled labour.** Shortages of specialists or skilled operators impact quality and on-time delivery to customers.

• **Resource utilisation.** Low utilisation of factory resources significantly impacts day-to-day output and OpEx. Effective use of manpower, machines, materials, and factory space is critical.

• **Sustainability.** This is now a non-negotiable part of any industrial strategy. Commitment to sustainable practices is essential from a compliance perspective.

The transformation

When we ask what smart manufacturing truly means for the electronics industry, the answer begins with recognising the factory challenges we face and identifying cost-effective technology solutions to solve them. From management's view, competitive pressures, operational inefficiencies, economic constraints,

and shifting market demands all push manufacturers toward digital transformation. Technologies such as AI, big data, and connected systems serve as foundational enablers.

• Connecting AI with smart products drives predictive capabilities

• Linking AI with big data enables advanced analytics

• Integrating big data with connectivity results in real-time data capture and monitoring

Yet, transformation is not limited to machines and technology. Operators on the shop floor play a pivotal role. Without proper training and mindset change, even the most advanced technologies cannot deliver expected outcomes. Thus, smart manufacturing rests on three pillars: process, technology, and people.

The transformation journey begins with consulting and extends across the smart manufacturing cloud. This journey can be visualised in three progressive stages:

1. Digital manufacturing.

Focuses on connected machines and factories. Connected machines require a communication architecture. Non-communicable and legacy machines need retrofitting with sensors and PLCs to become communicable.

2. **Digital factory.** Focus shifts to system Integration, using MES (manufacturing execution systems), ERP (enterprise resource planning), and PLM (product lifecycle management). Seamless interaction among these layers enables a truly connected environment.

3. **AI-driven manufacturing.** Captured data is leveraged for data-driven decision-making, supporting quality control and strategic business planning. Tools like machine vision and AI-powered predictive maintenance enhance production, ensure quality, and improve operational reliability.

From a data-driven perspective, these three phases are also known as Connect, Collect, and Consume.